Discrete mathematics I midterm test sample, autumn 2025

Solutions

1 a, (i)

$$\forall x\big(L(x) \Rightarrow \exists y(F(x,y) \wedge C(y))\big) \wedge \neg\big(\forall x\big(L(x) \Rightarrow \big(\forall y(F(x,y) \Rightarrow C(y))\big)\big)\big)$$

$$\Longleftrightarrow$$

$$\forall x\big(L(x) \Rightarrow \exists y(F(x,y) \wedge C(y))\big) \wedge \neg\big(\forall x\big(\forall y\big((L(x) \wedge F(x,y)) \Rightarrow C(y)\big)\big)\big)$$

$$\Longleftrightarrow$$

$$\forall x\big(L(x) \Rightarrow \exists y(F(x,y) \wedge C(y))\big) \wedge \big(\exists x\big(\neg\big(\forall y\big((L(x) \wedge F(x,y)) \Rightarrow C(y)\big)\big)\big)\big)$$

$$\Longleftrightarrow$$

$$\forall x\big(L(x) \Rightarrow \exists y(F(x,y) \wedge C(y))\big) \wedge \big(\exists x\big(\exists y\big(\neg\big((L(x) \wedge F(x,y)) \Rightarrow C(y)\big)\big)\big)\big)$$

$$\Longleftrightarrow$$

$$\forall x\big(L(x) \Rightarrow \exists y(F(x,y) \wedge C(y))\big) \wedge \big(\exists x\big(\exists y\big(L(x) \wedge F(x,y) \wedge \neg C(y)\big)\big)\big)$$

(any one of the above formulas is a correct answer on its own.)

1 a) (ii)

$$\neg\Big(\forall x\big(L(x)\Rightarrow\exists y\big(F(x,y)\wedge C(y)\big)\big)\wedge\big(\exists x\big(\exists y\big(L(x)\wedge F(x,y)\wedge\neg C(y)\big)\big)\Big)$$

De Morgan

$$\Longleftrightarrow$$

$$\neg\Big(\forall x\big(L(x)\Rightarrow\exists y\big(F(x,y)\wedge C(y)\big)\big)\Big)\vee\neg\Big(\exists x\big(\exists y\big(L(x)\wedge F(x,y)\wedge\neg C(y)\big)\big)\Big)$$

$$\Longleftrightarrow$$

$$\exists x\Big(\neg\big(L(x)\Rightarrow\exists y\big(F(x,y)\wedge C(y)\big)\big)\Big)\vee\Big(\forall x\Big(\neg\big(\exists y\big(L(x)\wedge F(x,y)\wedge\neg C(y)\big)\big)\Big)\Big)$$

$$\Longleftrightarrow$$

$$\exists x\Big(L(x)\wedge\neg\big(\exists y\big(F(x,y)\wedge C(y)\big)\big)\Big)\vee\Big(\forall x\Big(\forall y\Big(\neg\big(L(x)\wedge F(x,y)\wedge\neg C(y)\big)\Big)\Big)\Big)$$

De Morgan

$$\Longleftrightarrow$$

$$\exists x\Big(L(x)\wedge\big(\forall y\big(\neg\big(F(x,y)\wedge C(y)\big)\big)\big)\Big)\vee\Big(\forall x\big(\forall y\big(\neg L(x)\vee\neg F(x,y)\vee C(y)\big)\big)\Big)$$

De Morgan

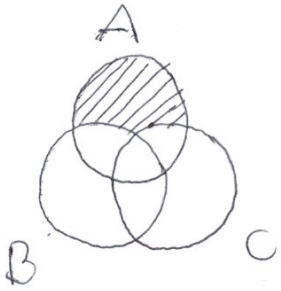
$$\Longleftrightarrow$$

$$\exists x\Big(L(x)\wedge\big(\forall y\big(\neg F(x,y)\vee\neg C(y)\big)\big)\Big)\vee\Big(\forall x\big(\forall y\big(\neg L(x)\vee\neg F(x,y)\vee C(y)\big)\big)\Big)$$

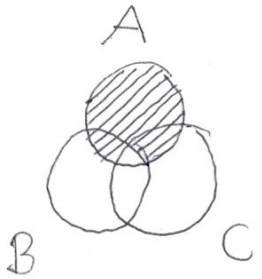
(any one of the 2nd, 3rd, 4th, 5th or 6th formulas above is a correct and complete answer on its own.)

1, b)

(i)



$A \setminus (B \cup C)$



$(A \setminus B) \cup (A \setminus C)$

⇒ <u>No</u>, the two sets are not always equal.

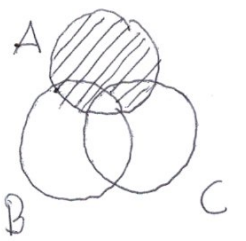
<u>Proof</u>: Counterexample: $A = B = \{1\}$, $C = \{2\}$.

$LHS = \{1\} \setminus (\{1\} \cup \{2\}) = \{1\} \setminus \{1,2\} = \emptyset$ ✗

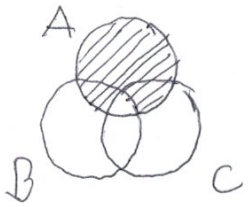
$RHS = (\{1\} \setminus \{1\}) \cup (\{1\} \setminus \{2\}) = \emptyset \cup \{1\} = \{1\}$

⇒ $LHS \neq RHS$ □

(ii)



$A \setminus (B \cap C)$



$(A \setminus B) \cup (A \setminus C)$

⇒ <u>Yes</u>, the two sets are always equal.

<u>Proof</u>:

$LHS = A \setminus (B \cap C) = A \cap (\overline{B \cap C}) = A \cap (\overline{B} \cup \overline{C})$

$RHS = (A \setminus B) \cup (A \setminus C) = (A \cap \overline{B}) \cup (A \cap \overline{C}) = A \cap (\overline{B} \cup \overline{C})$

⇒ $LHS = RHS$ □

2, a)

$dmn(S) = \{1,2,3,4,5,6\}$ $\quad mg(R) = \{1,2,4,5,6,7\}$

$\Rightarrow dmn(S) \triangle mg(R) = (dmn(S) \setminus mg(R)) \cup (mg(R) \setminus dmn(S)) =$

$= (\{1,2,3,4,5,6\} \setminus \{1,2,4,5,6,7\}) \cup (\{1,2,4,5,6,7\} \setminus \{1,2,3,4,5,6\}) =$

$= \{3\} \cup \{7\} = \underline{\underline{\{3,7\}}}$

b) $R(\{0,1,4,5,6,8\}) = \underline{\underline{\{4,5,1,6\}}}$

c) $R^{-1} = \{(4,1),(5,1),(7,2),(2,3),(1,6),(6,6),(4,7),(5,7),(7,7)\}$

$R^{-1}|_{\{2,4,5,7\}} = \underline{\underline{\{(4,1),(5,1),(7,2),(2,3),(4,7),(5,7),(7,7)\}}}$

d) $S \circ R = ?$ $\quad (x,z) \in S \circ R \Leftrightarrow \exists y : (x,y) \in R \wedge (y,z) \in S$

$(x,y) \in R \wedge (y,z) \in S$		$\Rightarrow (x,z) \in S \circ R$
$(1,4)$	$(4,1),(4,6)$	$\Rightarrow (1,1),(1,6)$
$(1,5)$	$(5,3),(5,4),(5,5)$	$\Rightarrow (1,3),(1,4),(1,5)$
$(3,2)$	$(2,1)$	$\Rightarrow (3,1)$
$(6,1)$	$(1,4)$	$\Rightarrow (6,4)$
$(6,6)$	$(6,6)$	$\Rightarrow (6,6)$
$(7,4)$	$(4,1),(4,6)$	$\Rightarrow (7,1),(7,6)$
$(7,5)$	$(5,3),(5,4),(5,5)$	$\Rightarrow (7,3),(7,4),(7,5)$

$\Rightarrow S \circ R = \underline{\underline{\{(1,1),(1,6),(1,3),(1,4),(1,5),(3,1),(6,4),(6,6),}}$

$\underline{\underline{(7,1),(7,6),(7,3),(7,4),(7,5)\}}}$

3, a) $\rho = \{(x,y) \in \mathbb{Z} \times \mathbb{Z} \mid 2x+1 = 3y-5\}$

$\rho(\{-1,0,1,2,3,4\}) = \{2,4\}$

$x=-1: \; 2(-1)+1 = 3y-5 \iff y = \frac{4}{3} \notin \mathbb{Z}$

$x=0: \; 2\cdot 0+1 = 3y-5 \iff y = 2 \in \mathbb{Z}$ ✓

$x=1: \; 2\cdot 1+1 = 3y-5 \iff y = \frac{8}{3} \notin \mathbb{Z}$

$x=2: \; 2\cdot 2+1 = 3y-5 \iff y = \frac{10}{3} \notin \mathbb{Z}$

$x=3: \; 2\cdot 3+1 = 3y-5 \iff y = 4 \in \mathbb{Z}$ ✓

$x=4: \; 2\cdot 4+1 = 3y-5 \iff y = \frac{14}{3} \notin \mathbb{Z}$

$\rho^{-1}(\{-1,0,1,2,3,4\}) = \{-3,0,3\}$

$y=-1: \; 2x+1 = 3(-1)-5 \iff x = -\frac{9}{2} \notin \mathbb{Z}$

$y=0: \; 2x+1 = 3\cdot 0-5 \iff x = -3 \in \mathbb{Z}$ ✓

$y=1: \; 2x+1 = 3\cdot 1-5 \iff x = -\frac{3}{2} \notin \mathbb{Z}$

$y=2: \; 2x+1 = 3\cdot 2-5 \iff x = 0 \in \mathbb{Z}$ ✓

$y=3: \; 2x+1 = 3\cdot 3-5 \iff x = \frac{3}{2} \notin \mathbb{Z}$

$y=4: \; 2x+1 = 3\cdot 4-5 \iff x = 3 \in \mathbb{Z}$ ✓

3, b) $R = \{(x,y) \in \mathbb{R}\times\mathbb{R} \mid 6x-1 = \frac{4y-3}{2}\}$

$S = \{(x,y) \in \mathbb{R}\times\mathbb{R} \mid y^2+2 = 7x\}$

$(x,z) \in \overset{②}{S} \circ \overset{①}{R} \iff \exists y : (x,y) \in R \wedge (y,z) \in S$

$\iff \exists y \in \mathbb{R}:$ ① $6x-1 = \frac{4y-3}{2}$ and ② $z^2+2 = 7y$

Expressing y from ②: $y = \frac{z^2+2}{7}$

Subst. $\frac{z^2+2}{7}$ for y in ①:

$$6x-1 = \frac{4\left(\frac{z^2+2}{7}\right)-3}{2} = 2\left(\frac{z^2+2}{7}\right) - \frac{3}{2} =$$

$$= \frac{2}{7}z^2 + \frac{4}{7} - \frac{3}{2} = \frac{2}{7}z^2 - \frac{13}{14}$$

$\iff 6x-1 = \frac{2}{7}z^2 - \frac{13}{14} \iff 84x - 14 = 4z^2 - 13$

$\iff 84x = 4z^2 + 1$

$\underline{S \circ R = \{(x,z) \in \mathbb{R}\times\mathbb{R} \mid 84x = 4z^2+1\}}$

4 a) $R = \{(1,1),(1,2),(1,3),(2,1),(2,2),(2,4),(3,1),(3,3),(4,2)\}$

on $X = \{1,2,3,4\}$.

REFL: No: $(4,4) \notin R$.

SYMM: Yes: $\forall x,y \in X : (x,y) \in R \Rightarrow (y,x) \in R$:

$(1,2) \in R \Leftrightarrow (2,1) \in R$ ✓✓

$(1,3) \in R \Leftrightarrow (3,1) \in R$ ✓✓

$(2,4) \in R \Leftrightarrow (4,2) \in R$ ✓✓

TRANS: No: $(1,2) \in R \wedge (2,4) \in R \Rightarrow (1,4) \in R$ is false, since $(1,4) \notin R$.

ANTI-SYMM: No: $(2,1) \in R \wedge (1,2) \in R$, where $1 \neq 2$.

b) $A = \{1,2,3,4,5\}$

$R = \{(1,2),(2,3),(2,1),(4,1),(4,5)\}$

5 a)

(i) $R_1 = \{(x,y) \in \mathbb{R} \times \mathbb{R} \mid xy \geq 0\}$ on $\mathbb{R}$

R_1 is not transitive, since e.g.:

$(-1,0) \in R_1$ (since $(-1)\cdot 0 = 0 \geq 0$)

and

$(0,1) \in R_1$ (since $0 \cdot 1 = 0 \geq 0$)

but

$(-1,1) \notin R_1$ (since $(-1)\cdot 1 = -1 \not\geq 0$).

Therefore <u>R_1 is not an equivalence relation.</u>

(ii) $R_2 = \{(x,y) \in \mathbb{Z} \times \mathbb{Z} \mid 4 \mid x-y\}$ on $\mathbb{Z}$.

<u>REFL</u>: yes: $\forall x \in \mathbb{Z}: 4 \mid \overbrace{(x-x)}^{0}$ (since $0 = 4 \cdot 0$),

hence $(x,x) \in R_2$.

<u>SYMM</u>: yes: $\forall x,y \in \mathbb{Z}$:

$(x,y) \in R_2 \Leftrightarrow 4 \mid x-y \Leftrightarrow \exists k \in \mathbb{Z}: x-y = 4k \Leftrightarrow$

$\Leftrightarrow y-x = -(x-y) = -4k = 4\underbrace{(-k)}_{\in \mathbb{Z}} \Rightarrow 4 \mid y-x \Leftrightarrow$

$\Leftrightarrow (y,x) \in R_2$.

<u>TRANS</u>: yes: $\forall x,y,z \in \mathbb{Z}$:

$(x,y) \in R_2 \wedge (y,z) \in R_2 \Leftrightarrow 4 \mid x-y \wedge 4 \mid y-z \Leftrightarrow$

$\Leftrightarrow \exists k_1, k_2 \in \mathbb{Z}: x-y = 4k_1 \wedge y-z = 4k_2 \Rightarrow$

$\Rightarrow \exists k_1, k_2 \in \mathbb{Z}: x-z = (x-y)+(y-z) = 4k_1 + 4k_2 =$

$= 4\underbrace{(k_1+k_2)}_{\in \mathbb{Z}} \Rightarrow 4 \mid x-z \Leftrightarrow (x,z) \in R_2$.

$\Rightarrow$ <u>yes, R_2 is an equivalence relation.</u>

5, b) $R_2 = \{(x,y) \in \mathbb{Z}\times\mathbb{Z} \mid 4 | x-y\}$ on $\mathbb{Z}$

Equivalence classes of R_2:

$[0] = \{y \in \mathbb{Z} \mid (y,0) \in R_2\} = \{y \in \mathbb{Z} \mid 4 | (y-0)\} = \{\ldots, -8, -4, 0, 4, 8, \ldots\} =$

$= 4\mathbb{Z}$

$[1] = \{y \in \mathbb{Z} \mid (y,1) \in R_2\} = \{y \in \mathbb{Z} \mid 4 | (y-1)\} = \{\ldots, -7, -3, 1, 5, 9, \ldots\} =$

$= \{4k+1 \mid k \in \mathbb{Z}\}$.

$[2] = \{y \in \mathbb{Z} \mid (y,2) \in R_2\} = \{y \in \mathbb{Z} \mid 4 | (y-2)\} = \{\ldots, -6, -2, 2, 6, 10 \ldots\} =$

$= \{4k+2 \mid k \in \mathbb{Z}\}$

$[3] = \{y \in \mathbb{Z} \mid (y,3) \in R_2\} = \{y \in \mathbb{Z} \mid 4 | (y-3)\} = \{\ldots, -5, -1, 3, 7, 11 \ldots\} =$

$= \{4k+3 \mid k \in \mathbb{Z}\}$

From here the equivalence classes start repeating:

$[4] = \{y \in \mathbb{Z} \mid (y,4) \in R_2\} = \{y \in \mathbb{Z} \mid 4 | (y-4)\} = \{\ldots, -8, -4, 0, 4, 8, \ldots\} =$

$= \{4k+4 \mid k \in \mathbb{Z}\} = \{4k \mid k \in \mathbb{Z}\} = 4\mathbb{Z} = [0]$.

$[5] = [1]$, $[6] = [2]$, ...

$\Rightarrow$ Four distinct equivalence classes:

$4\mathbb{Z}$, $\{4k+1 \mid k \in \mathbb{Z}\}$, $\{4k+2 \mid k \in \mathbb{Z}\}$, $\{4k+3 \mid k \in \mathbb{Z}\}$

$\Rightarrow$ Partition determined by R_2:

$\mathbb{Z}/R_2 = \{4\mathbb{Z}, \{4k+1 \mid k \in \mathbb{Z}\}, \{4k+2 \mid k \in \mathbb{Z}\}, \{4k+3 \mid k \in \mathbb{Z}\}\}$

6.) Version 1:

A relation $R \subseteq X \times X$ is a partial order if it is reflexive, transitive and anti-symmetric.

a) REFL: yes: $(1,1) \in T_1 \wedge (2,2) \in T_1 \wedge (3,3) \in T_1 \wedge (4,4) \in T_1$

TRANS: yes: $\forall x,y,z \in X : (x,y) \in T_1 \wedge (y,z) \in T_1 \Rightarrow (x,z) \in T_1$.

$(x,y) \in T_1 \wedge (y,z) \in T_1 \Rightarrow (x,z) \in T_1$

$(1,3) \quad (3,4) \Rightarrow (1,4)$ ✓

$(2,3) \quad (3,4) \Rightarrow (2,4)$ ✓

(In the above verification process I skipped those cases when $x=y$ or $y=z$, since in those cases $(x,z) = (y,z)$ or $(x,z) = (x,y)$, and hence in such cases the implication is trivially true.)

ANTI-SYMM: yes: $\forall x,y \in X : (x,y) \in T_1 \wedge x \neq y \Rightarrow (y,x) \notin T_1$

$(x,y) \in T_1 \wedge x \neq y \Rightarrow (y,x) \notin T_1$

$(1,3) \in T_1 \Rightarrow (3,1) \notin T_1$ ✓

$(1,4) \in T_1 \Rightarrow (4,1) \notin T_1$ ✓

$(2,3) \in T_1 \Rightarrow (3,2) \notin T_1$ ✓

$(2,4) \in T_1 \Rightarrow (4,2) \notin T_1$ ✓

$(3,4) \in T_1 \Rightarrow (4,3) \notin T_1$ ✓

$\Rightarrow$ T_1 is a partial order. □

6, Version 1

b, $T_2 = \{(x,y) \in \mathbb{Z} \times \mathbb{Z} \mid 5 \mid x-y\}$ on $\mathbb{Z}$.

T_2 is not anti-symmetric, since e.g.:

$(5,0) \in T_2$ (as $5 \mid (\overset{5}{5-0})$) and

$(0,5) \in T_2$ (as $5 \mid (\overset{-5}{0-5})$)

but $0 \neq 5$.

$\Rightarrow$ T_2 is not a partial order.

c, $T_3 = \{(x,y) \in \mathbb{Z} \times \mathbb{Z} \mid x+y \geq 2x\}$ on $\mathbb{Z}$.

Solution 1: Notice that $\forall x,y \in \mathbb{Z}: x+y \geq 2x \iff y \geq x$.

Therefore T_3 equals the standard $\leq$ relation on $\mathbb{Z}$.

We know that $\leq$ on $\mathbb{Z}$ is a partial order, hence

T_3 is a partial order.

6.) **Version 1:**

c) $T_3 = \{(x,y) \in \mathbb{Z} \times \mathbb{Z} \mid x+y \geq 2x\}$ on $\mathbb{Z}$

Solution 2:

REFL: $\forall x \in \mathbb{Z}: x+x = 2x \geq 2x$ ✓ $\Rightarrow$ yes.

TRANS:

Suppose $x+y \geq 2x$ and $y+z \geq 2y$ for some $x,y,z \in \mathbb{Z}$.

Then:

$$\left.\begin{array}{l} x+y \geq 2x \Leftrightarrow y \geq x \\ y+z \geq 2y \Leftrightarrow z \geq y \end{array}\right\} \Rightarrow z \geq y \geq x \Rightarrow z \geq x \Rightarrow x+z \geq 2x \checkmark$$

$\Rightarrow$ yes.

ANTI-SYMM:

Suppose $x+y \geq 2x$ and $y+x \geq 2y$. Then:

$$\left.\begin{array}{l} x+y \geq 2x \Leftrightarrow y \geq x \\ y+x \geq 2y \Leftrightarrow x \geq y \end{array}\right\} x = y \checkmark \Rightarrow \text{yes}$$

$\Rightarrow$ T_3 is a partial order.

6) Version 2:

a)

(i) $f_1 = \{(a,c), (b,b), (d,c), (e,a)\}$

Since each first coordinate occurs only in one ordered pair, f_1 is a function.

$(\forall x, y_1, y_2 : (x, y_1) \in f_1 \wedge (x, y_2) \in f_1 \Rightarrow f_1 = f_2)$

(ii) $f_2 = \{(x,y) \in \mathbb{R} \times \mathbb{R} \mid x^3 - 1 = y^2 + 2\}$

Solution 1: Not a function, since e.g. $x = 2$ is related to two different y values:

$2^3 - 1 = y^2 + 2 \iff 5 = y^2 \iff y_1 = \sqrt{5}$ or $y_2 = -\sqrt{5}$.

Solution 2: $\forall x, y \in \mathbb{R}$:

$x^3 - 1 = y^2 + 2 \iff x^3 - 3 = y^2 \iff y_{1,2} = \pm\sqrt{x^3 - 3}$

Therefore every $x \in \mathbb{R}$ such that $x^3 - 3 > 0$, x is related to two y values: $y_1 = \sqrt{x^3 - 3}$ and $y_2 = -\sqrt{x^3 - 3}$.

$\Rightarrow$ f_2 is not a function.

6.) **Version 2:**

a) (iii) $f_3 = \{(x,y) \in \mathbb{R}\times\mathbb{R} \mid \frac{4x-6}{4} = \frac{|y|+3}{6}\}$

Solution 1: Not a function, since e.g. $x=3$ is related to more than one y values ($y_{1,2} = \pm 15$):

$$\frac{4\cdot 3-6}{2} = \frac{|y|+3}{6} \iff |y| = 15 \iff y_{1,2} = \pm 15$$

Solution 2: $\forall x,y \in \mathbb{R}$:

$$\frac{4x-6}{4} = \frac{|y|+3}{6} \iff \frac{2x-3}{2} = \frac{|y|+3}{6} \iff$$

$$\iff 6x-9 = |y|+3 \iff |y| = 6x-12 \iff$$

if $6x-12 \geq 0$

$$\iff y_1 = 6x-12 \quad \text{or} \quad y_2 = -6x+12.$$

Therefore, if $6x-12 > 0$ (i.e. $x > -2$) then x is related to more than one y values: $y_1 = 6x-12$ and $y_2 = -6x+12$.

$\Rightarrow$ f_3 is not a function.

6.) b) $f_1 = \{(a,c),(b,b),(d,c),(e,a)\} \subseteq Y \times Y$

where $Y = \{a,b,c,d,e,f\}$.

INJ: Not injective, since $(a,c) \in f_1 \wedge (d,c) \in f_1$.

SURJ: $img(f_1) = \{a,b,c\} \neq Y$

$\Rightarrow$ Not surjective

BIJ: Since f_1 is neither injective nor surjective, it is not bijective.